[7]:
```
"""5. Password Strength Checker

Scenario: Write a program that checks the strength of a password based on these rules:

Length >= 8 characters: Strong

Length between 5 and 7 characters: Medium

Length < 5 characters: Weak"""
password = input("Enter your password: ")

if len(password) >= 8:
    print("Strong Password")
elif len(password) >= 5:
    print("Medium Password")
else:
    print("Weak Password")
```

```
Enter your password:  jj9597
Medium Password
```

[8]:
```
"""6. Electricity Bill Calculator

Scenario: An electricity company charges its customers as follows:

First 100 units: $0.5 per unit

Next 100 units (101-200): $0.75 per unit

Above 200 units: $1 per unit


Task: Write a program that accepts the number of units consumed and calculates the total bill."""


units = int(input("Enter electricity units: "))

if units <= 100:
    bill = units * 0.5
elif units <= 200:
    bill = (100 * 0.5) + (units - 100) * 0.75
else:
    bill = (100 * 0.5) + (100 * 0.75) + (units - 200) * 1

print("Total Electricity Bill: $", bill)
```

```
Enter electricity units:  230
Total Electricity Bill: $ 155.0
```

[9]:
```
"""7. Eligibility for Loan Approval

Scenario: A bank approves loans based on these conditions:

Age should be between 21 and 60

Monthly income should be greater than or equal to $5000


Task: Write a program to check whether a person is eligible for the loan based on their age and monthly income."""


age = int(input("Enter your age: "))
income = float(input("Enter your monthly income: "))

if age >= 21 and age <= 60 and income >= 5000:
    print("Eligible for Loan")
else:
    print("Not Eligible for Loan")
```

```
Enter your age:  23
Enter your monthly income:  60000
Eligible for Loan
```

[ ]:
```
'''8. Temperature Alert System

Scenario: A weather monitoring system alerts based on the temperature:

Below 0°C: "Freezing weather"

0°C to 20°C: "Cold weather"

21°C to 30°C: "Warm weather"

Above 30°C: "Hot weather"'''
```

[11]:
```

temp = float(input("Enter the temperature in Celsius: "))

if temp < 0:
    print("Freezing weather")
elif temp <= 20:
    print("Cold weather")
elif temp <= 30:
    print("Warm weather")
else:
    print("Hot weather")
```

```
Enter the temperature in Celsius:  12
Cold weather
```

[12]:
```
"""11. Leap Year Checker
Accept a year. First check whether it is divisible by 4. If yes, check whether it is divisible by 100.
If it is, check whether it is divisible by 400. Display whether the year is a leap year.

Divisible by 4 → Usually a leap year.
Divisible by 100 → Not a leap year.
Divisible by 400 → Leap year."""


year = int(input("Enter a year"))
if(year%4==0):
    if(year%100==0):
        if(year%400==0):
            print("leap year")
        else:
            print("not a leap year")
    else:
        print("not a leap year")
else:
    print("not a leap year")
```

```
Enter a year 2000
leap year
```

[13]:
```
"""
12. Triangle Classification
Accept three side lengths. First check whether they can form a valid triangle.
If valid, check whether all three sides are equal, exactly two sides are equal, or all sides are different.
"""



a = int(input("Enter first side: "))
b = int(input("Enter second side: "))
c = int(input("Enter third side: "))

if a <= 0 or b <= 0 or c <= 0:
    print("Invalid Triangle")
elif a + b <= c or a + c <= b or b + c <= a:
    print("Invalid Triangle")
elif a == b and b == c:
    print("Equilateral Triangle")
elif a == b or b == c or a == c:
    print("Isosceles Triangle")
else:
    print("Scalene Triangle")
```

```
Enter first side:  3
Enter second side:  2
Enter third side:  4
Scalene Triangle
```